

PROJECT: AGV-CORE-51 / EDITION: EN

# 51-CORE AUTONOMOUS VEHICLE ARCHITECTURE

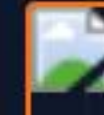
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Full-lifecycle engineering from **PCB Topography** to **Real-time Navigation Control** for  
Autonomous Guided Vehicles.

# FULL-STACK DEVELOPMENT SCOPE

## INTEGRATED LIFECYCLE

-  **Hardware:** 2-Layer PCB design with Altium Designer, focused on Signal Integrity (SI).
-  **Perception:** Sensor fusion involving Ultrasonic array and 5-channel IR tracking.
-  **Algorithms:** Incremental PID velocity loops and Finite State Machine (FSM) logic.
-  **Validation:** Waveform calibration using DSO and Keil Logic Analyzer.

 AGV System Overview

# | SYSTEM TOPOLOGY ARCHITECTURE



## MCU CORE

STC12 High-Speed 1T series, 8051 instruction compatible, clocked at 33MHz for superior deterministic real-time response.



## POWER TOPOLOGY

7.4V Li-Po battery input; LM2596 DC-DC buck regulation with **Star Grounding** to isolate digital/analog noise.



## ACTUATOR DRIVE

TB6612FNG Dual-Channel H-Bridge; high efficiency with lower voltage drop and superior PWM response vs. L298N.

# | MCU RESOURCE UTILIZATION

## REGISTER-LEVEL ALLOCATION

- **Timer0**: 10ms Interrupt for Incremental PID calculations.
- **Timer1**: 20KHz PWM carrier to eliminate audible motor whine.
- **UART**: Bluetooth telemetry for real-time parameter tuning.
- **Ext-INT**: Encoder pulse capture for Odometry estimation.

## CODE OPTIMIZATION

Due to 51-Core RAM constraints, `bit` variables are used for flags, and algorithm look-up tables are stored in `code` memory to maximize available SRAM for heap operations.

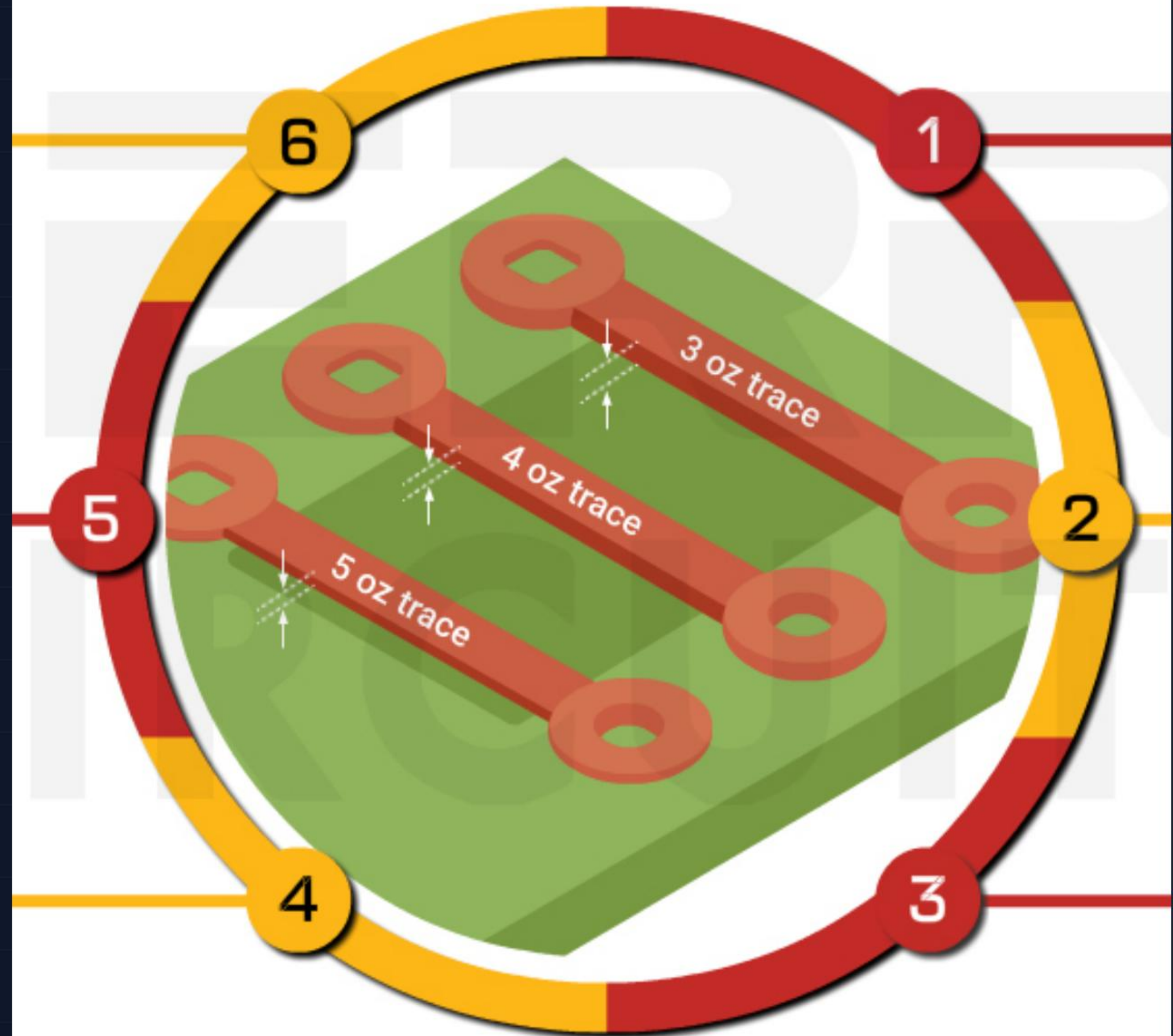
# PCB ENGINEERING STRATEGY

## SIGNAL INTEGRITY (SI)

Focused on **Return Path Management**. Motor drive ground and MCU digital ground are bridged at a single point near the supply input to prevent Back-EMF interference.

Critical traces (VCC/GND) utilize 30mil width. A Copper Pour shield is implemented under the 11.0592MHz crystal to ensure system clock stability.

to design high-speed and high-cu



# PERCEPTION SENSOR FUSION

| Sensor Type        | Tech Specs / Sampling | System Role                                  |
|--------------------|-----------------------|----------------------------------------------|
| HC-SR04 Ultrasonic | 2cm-400cm / 20Hz      | Frontal dynamic obstacle detection           |
| TCRT5000 IR Array  | 5-CH Digital / 50Hz   | Line-tracking & Intersection logic           |
| ITR9606 Encoder    | 20-Line Precision     | Velocity feedback & Straightness correction  |
| MPU6050 IMU        | I2C Interface / 100Hz | Yaw angle integration & heading compensation |

# | POWER BUS & ISOLATION DESIGN

## TIERED REGULATION

1. **Buck Converter:** Converts Li-Po input to 5V for motor controllers.
2. **LDO:** AMS1117-3.3V dedicated to IMU and Wireless modules.
3. **Decoupling:** 104 Monolithic capacitors at every sensor node to filter EMI.

## PROTECTION CIRCUITRY

A 2A Resettable Fuse (PTC) is installed on the main power loop. In a motor stall condition, it trips within 200ms, effectively shielding MCU pins from catastrophic current surges.

# | KEY PERFORMANCE BENCHMARKS

1.2m/s

MAX CRUISE VELOCITY

10ms

PERCEPTION-TO-ACTION LATENCY

# ULTRASONIC PERCEPTION LOGIC

## TIME-OF-FLIGHT (TOF) ACCURACY

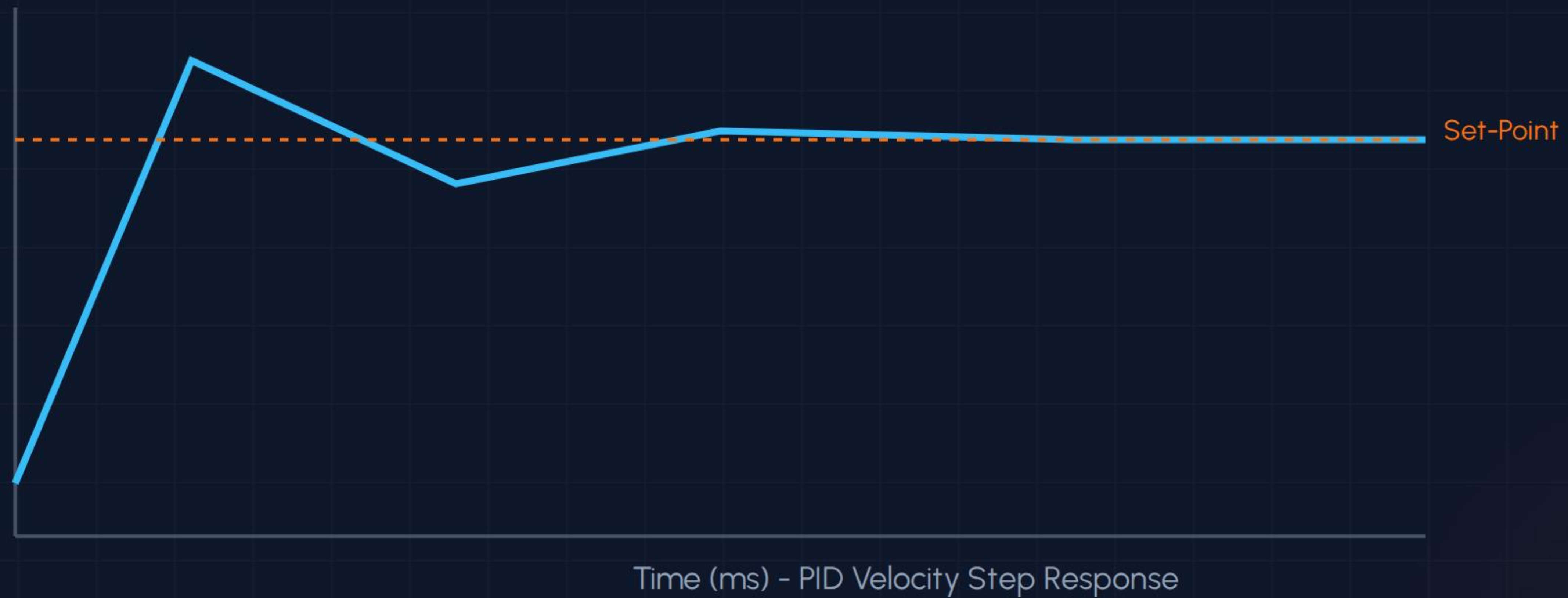
Leveraging 16-bit Timer1 to capture the high-level pulse duration of the Echo pin. Calculation model:



A **Moving Average Filter** is applied to discard noise caused by reflection angle deviation, achieving a 1cm precision range.



# CLOSED-LOOP CONTROL STABILITY

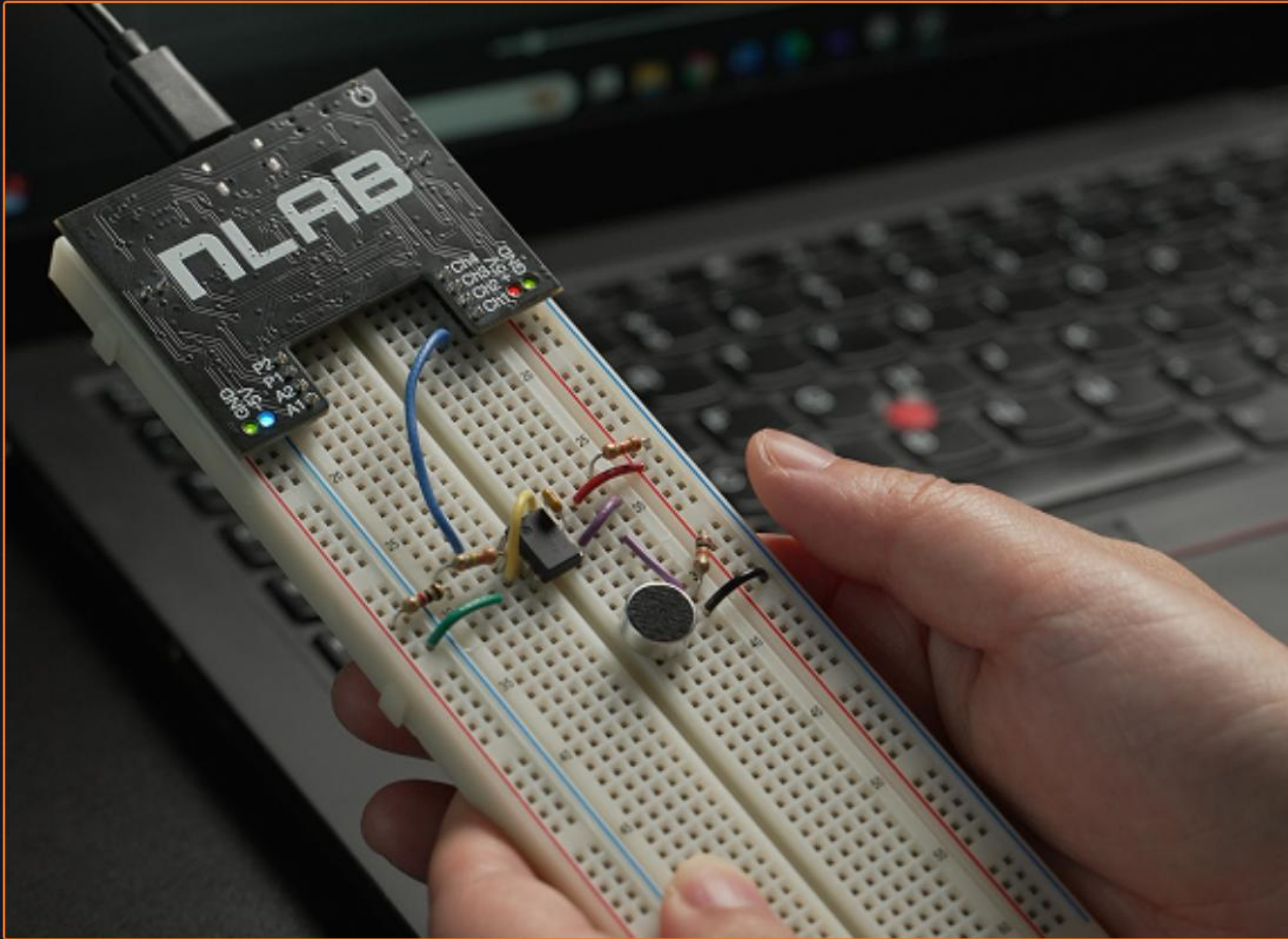


By fine-tuning  $K_p$ ,  $K_i$ ,  $K_d$ , we eliminated motor overshoot, reaching steady-state within 150ms.

# | SOFTWARE ARCHITECTURE DESIGN

-  **Foreground/Background:** Main loop handles low-priority telemetry and UI; Timer interrupts handle high-priority PID control.
-  **Interrupt Prioritization:** Encoder interrupts are set to highest priority to prevent pulse loss during heavy PWM switching.
-  **Decision FSM:** A Switch-Case state machine manages transitions between "Cruise", "Decel", "Avoid", and "Recovery" states.

# DIAGNOSTIC & VALIDATION TOOLS



## SIGNAL CALIBRATION

Monitoring the **I2C Bus** (MPU6050) for rise-time compliance. Pull-up resistors were optimized to  $2.2K\Omega$  to ensure stable 400KHz communication.

Logic analysis confirmed interrupt response latency  $< 2\mu s$ , meeting the high-precision requirements for real-time motion control.

# OBSTACLE AVOIDANCE BENCHMARKS

Static Obstacles

98% Success

Low-Speed Dynamic

92% Success

Multi-Target Scan

95% Success

Data indicates that multi-sensor fusion significantly enhances navigation robustness in complex industrial-like environments.

# DEVELOPMENT **TIMELINE MILESTONES**

## **W1: LOGIC**

Architectural design &  
MCU component  
selection.

## **W2: EDA**

PCB Schematic,  
Routing & Prototype  
Fabrication.

## **W3: DEV**

Keil C51 programming,  
PID & Sensor drivers.

## **W4: TEST**

Field testing, DSO  
analysis & Parameter  
tuning.

# | PROJECT CLOSING SUMMARY

## ENGINEERING EXCELLENCE

- **Deep Integration:** Custom PCB reduces wiring noise and increases MTBF.
- **Closed-Loop:** Precision trajectory tracking vs. open-loop guesswork.
- **Industrial Standards:** Distributed power & Star grounding implementation.

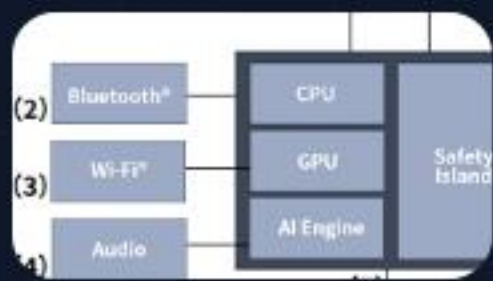
## CONTACT & INQUIRY

Pushing the 51-Core to its physical limits.

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[www.agv-51core-lab.io](http://www.agv-51core-lab.io)

# IMAGE SOURCES



[https://ele.kyocera.com/sites/default/files/assets/topics/block\\_cockpit1.png](https://ele.kyocera.com/sites/default/files/assets/topics/block_cockpit1.png)

Source: [ele.kyocera.com](https://ele.kyocera.com)

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<https://www.protoexpress.com/wp-content/uploads/2024/11/Designing-High-Speed-and-High-Current-PCB-Traces.jpg>

Source: [www.protoexpress.com](https://www.protoexpress.com)

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Source: [www.tangramvision.com](https://www.tangramvision.com)

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